

11 (long)

(a) Under fatigue loading, a crack in a gas turbine blade made of titanium alloy grows at a rate

$$\frac{da}{dN} = 4 \times 10^{-11} \Delta K^5 \text{ (m cycle}^{-1}\text{)}$$

where a is the crack length, N is the number of fatigue cycles and ΔK is the stress intensity factor range (measured in units of $\text{MPa m}^{1/2}$). The fracture toughness of the titanium alloy is $K_{IC} = 30 \text{ MPa m}^{1/2}$. In service, the blade experiences a constant stress range with a $\sigma_{\min} / \sigma_{\max}$ ratio of 0.1. During operation, the blade failed after 10 000 loading cycles. Analysis revealed that failure resulted from the growth of an internal circular fatigue crack. The crack had grown to a radius of 5 mm, before the component failed by fast fracture. The stress intensity factor K for an internal circular crack of radius a is given by $K = (2 / \pi) \sigma \sqrt{\pi a}$ where σ is the remote stress.

(i) Determine the maximum stress normal to the plane of the crack during a loading cycle, and the stress range of the cycle. [8]

(ii) What was the initial crack length? [10]

(b) The blades of a gas turbine are to be manufactured from silicon nitride. A blade is 30 mm long, it has a cross-sectional area of $20 \times 10^{-6} \text{ m}^2$ and is attached to a rotor of radius $R = 200 \text{ mm}$. When the rotor rotates at an angular velocity of $\omega \text{ rad s}^{-1}$, the axial stress $\sigma(x)$ in the blade may be approximated by

$$\sigma(x) = \rho \omega^2 R x$$

where x is the distance from the tip and $\rho = 3200 \text{ kg m}^{-3}$ is the density of the material.

In service, the rotor operates at a maximum speed of $\omega = 3000 \text{ rad s}^{-1}$. The design allows for a maximum failure probability of 10^{-7} . Assuming a Weibull modulus $m = 10$, specify a target uniaxial tensile strength which corresponds to a failure probability of 1% in tests conducted on samples of cross-sectional area $16 \times 10^{-6} \text{ m}^2$ and length 40 mm. Use the Weibull equation for the survival probability P_s from the Materials Data Book. [12]